

Younghyeon Park

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Research Interests

Video signal processing, machine learning, compressive sensing, imaging system design, multi-spectral imaging, light-field microscopy

Education

2011-2022 **Sungkyunkwan University** Ph.D. in Engineering Electrical and Computer Engineering (GPA: 4.24/4.5)
Thesis: A system design of computational visible and near-infrared imaging
Advisor: Byeungwoo Jeon

2004-2011 **Sungkyunkwan University** Bachelor Electronic and Electrical Engineering (GPA: 3.56/4.5)

Work Experience

Jan. 2026 – Present **Curi Bio, Co. Ltd. (KR)**, Director, Republic of Korea

- System design of light-field microscopy

Apr. 2020 – Dec. 2025 **Qubit Bio, Inc. (Acquired by Curi Bio, Inc.)**, Chief Technical Officer, Republic of Korea

- System design of light-field microscopy
- System design of antimicrobial susceptibility testing device

Jun. 2006 – Dec. 2008 **Bluebird Inc.**, Researcher, Republic of Korea

- Full-time work as researcher in platform development department
- Windows mobile OS porting and board support package design for customized hardware platform
- Device driver development for Audio-path and phone core interface between modem and OS
- Programming experience on C++ language and ARM assembly

Awards

Jan. 2018 Best poster presentation

- International conference on electronics information and communication (ICEIC2018)
- Title: Visible and Near-Infrared Separation using Conditional Generative Adversarial Network
- Role 1st author

Nov. 2010 1st place award

- Graduate exhibition, Sungkyunkwan University
- Title: Smart Oriented Driving in the U-City
- Role: Automated driving of multiple line-tracers by mean-shift algorithm to detect car position and Dijkstra's shortest path algorithm to find driving path

Aug. 2006 2nd place award

- 7th DSP design contest, Texas Instrument Korea Ltd.
- Title: Implementation of biped robot using TMS320LF2406A
- Role: Hardware design and walking analysis of biped robot

Professional Affiliations

2012-Present Member, IEEE Signal Processing Society

Tech Stack

Research Background: Computational photography, Compressive sensing, Image processing, Video Coding (HEVC, AVC, VP8, Distributed Video Coding)

Software Programming: C, C++, Java(Android), Matlab, Python, Verilog, VHDL, Assembly

Hardware design: Circuit & PCB design (OrCAD, Altium designer, KiCAD)

Languages: Korean, English

Publications

International Journal

- [1] S. Kaushal, S. Shrivastava, Y.-R. Yun, Y. Park, T. T.-Q. Nguyen, M. Meeseepong, E. Lee, B. Jeon, M. B. Gu, S. Yang, N.-E. Lee, "Culture-free quantification of bacteria using digital fluorescence imaging in a tunable magnetic capturing cartridge for onsite food testing," ACS sensors, Vol. 7, No. 8, pp. 2188-2197, Aug. 2022.
- [2] Y. Park, B. Jeon, "An Acquisition Method for Visible and Near Infrared Images from Single CMYK Color Filter Array-based Sensor," MDPI Sensors, Vol. 20, No. 19, pp. 5578, Sep. 2020.
- [3] W.-I. Lee, Y. Park, S. Shrivastava, T. Jung, M. Meeseepong, J. Lee, B. Jeon, S. Yang, N.-E. Lee, "A fully integrated bacterial pathogen detection system based on count-on-a-cartridge platform for rapid, ultrasensitive, highly accurate and culture-free assay," Biosensors and Bioelectronics, Vol. 152, Jan. 2020.
- [4] W.-I. Lee, Y. Park, J. Park, S. Shrivastava, Y.-M. Son, H.-J. Choi, J. Lee, B. Jeon, H. Lee, N.-E. Lee, "A smartphone fluorescence imaging-based mobile biosensing system integrated with a passive fluidic control cartridge for minimal user intervention and high accuracy," Lab on a Chip, Vol. 19, No. 8, pp. 1502-1511, Mar. 2019.
- [5] H. Lee, S. Yang, Y. Park, B. Jeon, "Fast Quantization Method with Simplified Rate-Distortion Optimized Quantization for HEVC Encoder," IEEE Transactions on Circuits and Systems for Video Technology, vol. 26, no. 1, pp. 107-116, Jan. 2016.
- [6] H. Lee, H. J. Shim, Y. Park, B. Jeon, "Early Skip Mode Decision for HEVC Encoder with Emphasis on Coding Quality," IEEE Transactions on Broadcasting, vol. 61, no. 3, pp. 388-397, Sep. 2015.
- [7] K. Q. Dinh, C. V. Trinh, V. A. Nguyen, Y. Park, B. Jeon, "Measurement Coding for Compressive Sensing of Color Images," IEEE Transactions on Smart Processing & Computing, vol. 3, no. 1, pp. 10-18, Feb. 2014.
- [8] H. Lee, Y. Park, B. Jeon, "Optimal complexity scalable H.264/AVC video decoding scheme for portable multimedia devices" Optical Engineering, vol. 52, no. 7, pp. 071508-1-11, July 2013.
- [9] Y. Park, H. Lee, B. Jeon, "Performance Analysis of Open Web Video Codec VP8," IEEE Transactions on Smart Processing & Computing, vol. 2, no. 2, pp. 86-96, Apr. 2013.

International Conference

- [1] E. Choi, Y. Park, B. Jeon, "A drone-projected image stabilization: consideration of tilt and scaling," International workshop on advanced imaging technology (IWAIT), Vol. 11515, pp. 579-584, June 2020.
- [2] E. Choi, Y. Park, B. Jeon, "An implementation of drone-projector: stabilization of projected image," IS&T International Symposium on Electronic Imaging, pp. 332-1~332-5, Jan. 2020.
- [3] Y. Park, B. Jeon, "Spatially High Resolution Visible and Near-Infrared Separation using Conditional Generative Adversarial Network and Color Brightness Transfer Method," The 10th International Conference on Communications, Circuits and Systems (ICCCAS), Dec. 2018.
- [4] J. Lee, Y. Park, B. Jeon, "Low intensity rgb texture enhancement based on near infrared image using perceptual information," The 10th International Conference on Communications, Circuits and Systems (ICCCAS), Dec. 2018.
- [5] Y. Park, B. Jeon, "Visible and near-infrared separation using conditional generative adversarial network," 2018 International Conference on Electronics, Information, and Communication (ICEIC), Hawaii, Jan. 2018.
- [6] Y. Park, H. J. Shim, K. Q. Dinh and B. Jeon, "Visible and near-infrared image separation from CMYK color filter array based sensor," 2016 International Symposium ELMAR, pp. 209-212., Zadar, Sep. 2016.
- [7] K. Q. Dinh, Y. Park, T. N. Canh, and B. Jeon, "Rate distortion optimization for omnidirectional video using HEVC," Global 3D Tech forum, pp. 38-39, Oct. 2016.

- [8] K. Won, Y. Park, B. Jeon, "Motion Vector Decision for Low Complexity HEVC Encoding," International SoC Design Conference, pp. 225-228, Busan, South Korea, Nov. 2013.
- [9] V. A. Nguyen, Y. Park, H. J. Shim, and B. Jeon, "Feature Adaptive BCS-SPL for Image", Proc. of IWAIT2013, Nagoya, Japan, Jan. 2013.

Patent

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- [1] B. Jeon, J. Lee, Y. Park, "Method of deblurring motion in light field image and device therefor," Sungkyunkwan University Research and Business Foundation, US 2024/0273687 A1 (Application), KR 10-2735759 (Registration).
- [2] B. Jeon, J. Lee, Y. Park, "Shutter array and light field imaging device including the same," Sungkyunkwan University Research and Business Foundation, KR 10-2023-0018107 (Application).
- [3] W. Lee, Y. Park, S. Hong, M. Meeseepong, "LENS MODULE, OPTICAL DEVICE AND 3D IMAGE COLLECTION METHOD USING THE SAME," Quvit Bio, Inc., KR 10-2022-0186543, (Application).
- [4] W. Lee, Y. Park, "LIGHT FIELD MICROSCOPY-BASED IMAGE ACQUISITION METHOD AND DEVICE," Quvit Bio, Inc., KR 10-2764645 (Registration), US 12598400B2 (Registration).
- [5] B. Jeon, Y. Park, J. Lee, "METHOD AND APPARATUS FOR MACHINE-LEARNING BASED PROCESSING OF HIGH-RESOLUTION IMAGE," Sungkyunkwan University Research and Business Foundation, KR 10-2599675, 11/02/2023 (Registration).
- [6] W. Lee, Y. Park, "APPARATUS AND METHOD FOR DISTINGUISHING ANTIBIOTICS SUSCEPTIBILITY," KR 10-2551052, 06/29/2023 (Registration).
- [7] K.-H. Won, C.-Y. Kim, B. Jeon, K. Q. Dinh, Y. Park, H.-J. Shim, S.-I. Lee, "Method and Apparatus for Outputting Images," Samsung Electronics Co., Ltd., Sungkyunkwan University Research and Business Foundation, US 10027933 B2, 01/17/2018 (Registration).
- [8] B. Jeon, K. Q. Dinh, Y. Park, T. N. Canh, "AN APPARATUS FOR COLOR IMAGE DENOISING AND METHOD THEREOF," Sungkyunkwan University Research and Business Foundation, KR 10-1791789, 10/24/2017 (Registration).
- [9] K.-H. Won, C.-Y. Kim, B. Jeon, K. Q. Dinh, Y. Park, H.-J. Shim, S.-I. Lee, "Apparatus for outputting image and method thereof," Samsung Electronics Co., Ltd., Sungkyunkwan University Research and Business Foundation, KR 10-2287944, 08/03/2021 (Registration).

Key research projects

2021-Present Single-shot 3D sample imaging via light-field technology

- Design of Fourier light-field microscopy
- Design of well-plate scanning based imaging system
- Design of GPU accelerated fast 3D reconstruction algorithm
- Supported by Korea Institute for Advancement of Technology (KIAT)

2020-2021 Antimicrobial Susceptibility Testing Device Development

- Implementation of 12ch pipetting system
- Design of fluorescence/phase-contrast imaging system
- Design of antimicrobial susceptibility testing software algorithm
- Supported by Korean government startup funding program

2018-2022 Computational Multispectral Video System Design using Coded Illumination

- Design of novel multispectral video camera for low light condition
- High speed acquisition of 16 band multispectral data and motion deblurring using coded illumination
- Supported by Institute for Information & communications Technology Promotion (IITP)

2016-2018 Visible and Near-Infrared (NIR) Image Separation Technique based on Deep Learning

- RGB, NIR separation from an image taken by CMYK color filter array based sensor
- RGB, NIR separation from an image taken by RGB color filter array based sensor using deep neural network
- Supported by National Research Foundation of Korea

2017-2019 Integrated Representation and Compression Technique for Multifunctional Image Data

- Develop a novel compression technique for multi-functional data (i.e. light-field or hyperspectral image/video)
- Function scalable compression for immersive media (e.g. transmission of light-field data for VR display)
- Supported by National Research Foundation of Korea

Other research activities

- Distributed video coding on mobile device
- Light-weight video coding based on compressive sensing
- Lens-less camera implementation